

Tutorial 4 Group 2

Desa Mentari Blok 5

Safe Flame, Green Aim

A Community Initiative to Extinguish
Fire Risks and Foster Environmental Well-being.



MPU 3422 Community Service for Planetary Health

Understanding Desa Mentari Blok 5

Introduction to the Host Community of Desa Mentari Blok 5

About the Desa Mentari Blok 5

About the Desa Mentari Community:

- Located approximately 3.3 KM away from Sunway City
- An affordable housing community that consist of 10 blocks of 11 story buildings

Population Dynamics and Cultural Diversity:

- A population density of between 26,000 to 27,000
- Multiracial community comprising of all three Malaysian races
- Individuals comprise mostly of the B40 socioeconomic group



The Desa Mentari Blok 5

*Sources

- Google Maps for location & no. of blocks
- Statistics are directly from the Desa Mentari Community Leaders

Past Fire Incidents at Desa Mentari

Fire Incidents Have Occurred as Recently as 2022

- According to Shariful Azni (2022), a fire incident had occurred on the 8th of May 2022.
- The fire on the 6th floor of Block A destroyed 80% of the apartment unit
- This **highlights the urgent need for improved fire safety measures**, such as better fire extinguishers, awareness programs, and maintenance regulations.



Above Left: The Damaged Unit of Block A



Above Right: The Fire at Block A of Desa Mentari

***Source:**

- **Sinar Harian:** (Shariful Azni, 2022)
<https://www.sinarharian.com.my/article/201304/BERITA/Semasa/Warga-Indonesia-terselamat-rumah-terbakar-80-peratus>

Understanding the Problem of Desa Mentari Blok 5

Understanding the Problem Statement and Problem Diagnosis

Problem Statement

The Desa Mentari community **faces a high fire safety risk** due to **insufficient emergency fire infrastructure, limited funding, and low community awareness**. These factors increase the likelihood of fire outbreaks, delayed response times, and potential loss of life or property. Vulnerable groups, such as children and the elderly, are particularly at risk, highlighting the urgent need for improved fire safety measures.

Problem Diagnosis

Main Problem	High fire safety risk in Desa Mentari due to insufficient emergency fire infrastructure.
Causes	• Limited funding and outdated maintenance practices. • Lack of community awareness and training on fire safety. • Absence of knowledge or experience in fire safety standards.
Key Stakeholders Involved	• Contributors: Building management, local authorities, contractors cutting costs. • Problem Solvers: Community leaders, residents, and university programs. • Negatively Affected: Residents, especially vulnerable groups, and local businesses. • Beneficiaries: Construction companies that may profit from land acquisition.

Initiatives to Enhance Fire Safety in Desa Mentari

Strategic Actions to Mitigate Fire Risks and Educate the Community

(1) Fire Safety Education and Awareness

- **Expert-Led Training Session:** Conducted a fire safety education workshop in collaboration with a certified fire officer to educate residents on fire prevention and emergency response.
- **Community Engagement:** Empowered local residents with actionable knowledge to reduce fire hazards.
- **Introduction to the P.A.S.S. Methodology:**
 - PULL the pin to unlock the extinguisher.
 - AIM the nozzle at the base of the fire.
 - SQUEEZE while pointing the nozzle downwards.
 - SWEEP by moving the nozzle side to side.



Fire Education Session by Mr. William



Fire Safety Standards Education by Mr. William

Initiatives to Enhance Fire Safety in Desa Mentari

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(2) Hands-On Fire Response Training

- **Live Fire Extinguishing Demonstration:** Organized a practical fire suppression session with local leaders, providing residents with real-world experience in handling fire emergencies.
- **Leadership Involvement:** Strengthened community preparedness by ensuring local leaders play an active role in emergency response training.
- **Simulated Fire Emergency Scenarios:** The Certified Fire Officer guided residents through various real-life fire situations, demonstrating how to assess the severity of a fire, determine whether to safely extinguish it or evacuate immediately, and follow proper emergency protocols to minimize risks.



The Education of Simulated Fire Emergency Scenarios



Live Fire Extinguishing Demonstration & Practical Training for Residents

Initiatives to Enhance Fire Safety in Desa Mentari

Strategic Actions to Mitigate Fire Risks and Educate the Community

(3) Strengthening Fire Safety Infrastructure

- **Deployment of Fire Extinguishers:** Installed 11 fire extinguishers, ensuring each floor in Blok 5 has access to fire suppression tools.
- **Targeted Risk Mitigation:** Addressed critical gaps in fire response readiness through improved infrastructure placement.
- **Strengthened Local Involvement:** Designated resident representatives on each floor of Block 5 to undergo hands-on practical training, equipping them with the necessary fire safety skills to act as first responders in emergencies and guide fellow residents during fire incidents.



Involvement of the local leaders of Desa Mentari



Distribution of Fire Extinguishers to Residents

Initiatives to Enhance Fire Safety in Desa Mentari

Strategic Actions to Mitigate Fire Risks and Educate the Community



Group Photo between Students and Local Residents and Community Leaders

Initiatives to Enhance Fire Safety in Desa Mentari

Key Deliverables Implemented to Protect the Desa Mentari Community



Empowering the Community:

Over 30 local residents equipped with essential fire safety knowledge and hands-on training to enhance emergency preparedness.



Proactive Fire Prevention:

Educated 30+ residents on essential fire prevention measures, with 85% reporting increased confidence in identifying and mitigating fire hazards.



Strengthening Fire Safety Infrastructure:

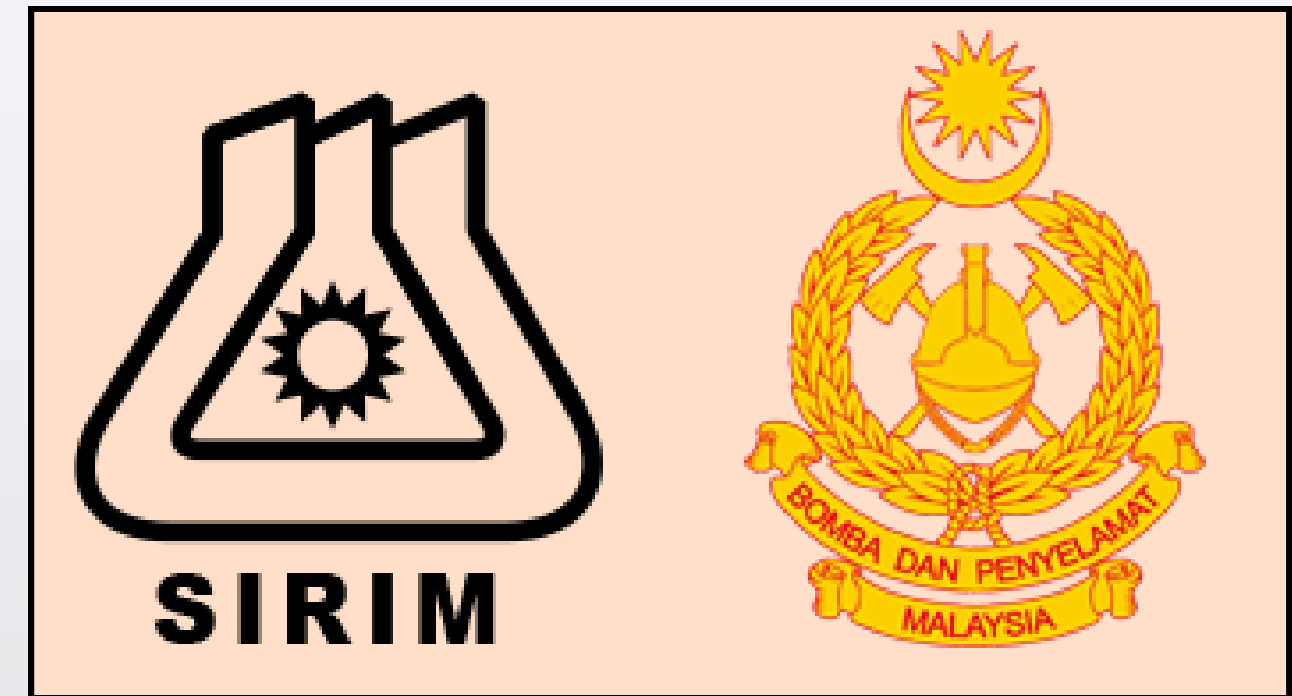
Distributed 11 fire extinguishers across all floors of Blok 5, ensuring 60% improvement in coverage for emergency response.

Challenges Faced in Project Planning & Execution

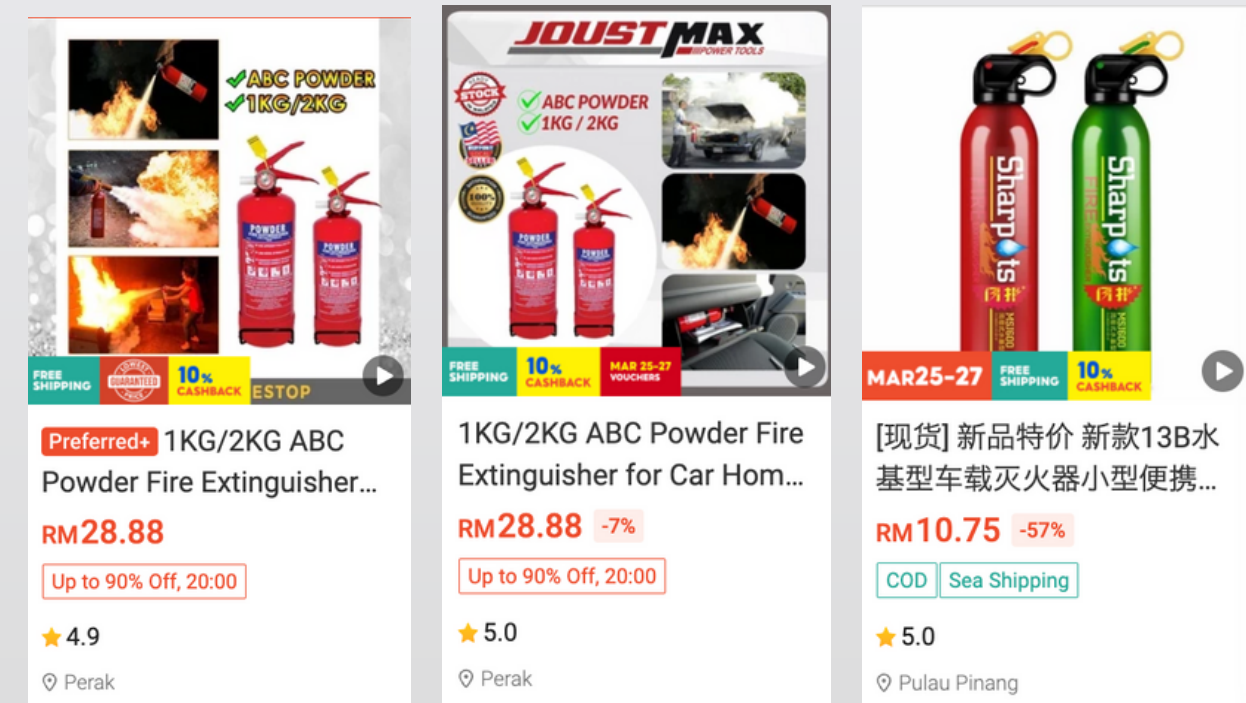
Challenge Encountered - Unapproved Fire Extinguishers

Challenges Finding Approved Fire Extinguishers

- Initial attempts to procure fire extinguishers via Shopee revealed that these units were not certified by BOMBA/SIRIM.
 - SIRIM certification** helps manufacturers ensure their products are safe and compliant with national standards (SIRIM QAS International Sdn. Bhd, 2022).
 - A **Bomba License** is a sticker affixed to fire extinguishers after they have been inspected and approved by Bomba officials (Palcon Engineering & Development, 2024)
- While these options were cheaper, their reliability and safety could not be guaranteed.



The above logos represent an approved fire extinguisher



Examples of cheaper non-licensed fire extinguishers

*Sources

- <https://www.palcon.com.my/fire-extinguisher-renewal-malaysia/>
- <https://www.sirim-qas.com.my/our-services/product-certification/fire-listing-scheme/>

Challenges Faced in Project Planning & Execution

Solution to Challenge - Combination of Crowd Funding and Direct Sourcing

Direct Sourcing from Supplier

- **Negotiated Bulk Discounts:** Purchasing directly from certified suppliers allowed for cost reductions through volume discounts.
- **Ensured Compliance:** Fire extinguishers and other safety equipment were sourced only from BOMBA/SIRIM-approved vendors to meet safety standards.
- **Challenges:** Due to the relatively small order size of 11 fire extinguishers, the bulk purchase discount was minimal, limiting cost savings. Despite this, the Official Fire Instructor provided the fire safety demonstration at no additional cost.



Directly sourced from Fire Fighter



Fire Fighter's Warehouse where the fire extinguishers were procured from

Potential Challenges & Strategic Mitigation Strategies

Optimizing Fire Safety Equipment for Long-Term Effectiveness

Challenge: Maintenance and Inspection

- Without periodic checks, 40% of fire extinguishers in similar community initiatives were found non-functional after one year.
- Risk of equipment failure in critical emergencies due to leaks, pressure loss, or expiration.

Strategic Mitigation Strategies

- Implement a quarterly inspection protocol: *100% of extinguishers must be checked every 3 months.*
- Assign one community leader per zone to oversee maintenance, ensuring zero expired units.

Challenge: Loss or Misuse of Equipment

- In similar projects, 20-30% of project deliverables were either lost or misused within the first 6 months, posing a risk of something similar occurring.
- Potential risk of zero availability during emergencies due to missing or discharged units due to negligence.

Strategic Mitigation Strategies

- Secure 100% of extinguishers with a specified resident for safeguarding.
- Conduct bi-annual community awareness workshops on proper usage, aiming for 80%+ resident participation to increase awareness.

Final Reflections & Lessons Learned

Key Takeaways from Planning & Implementation

- **Procurement Challenges:**
 - Bulk order discounts were minimal due to small order size (11 units).
 - Involvement of more groups with larger bulk orders could help drive down per unit cost.
- **Regulatory Compliance:**
 - Familiarity with the respective BOMBA/SIRIM-approved extinguishers could help ease the procurement process.
- **Stakeholder Engagement:**
 - Fire safety demonstration was secured at no additional cost, ensuring better training for residents.
 - Engagement with local community leaders helped ensure resident participation and attendance for the entire project.



The Involvement of Local Community Leaders



Official BOMBA / SIRIM approved fire extinguishers

Final Reflections & Lessons Learned

Impact on the Desa Mentari Community

- **Enhanced Fire Safety Awareness:**
 - Over 60% of participants gained hands-on experience in extinguisher usage.
- **Infrastructure Upgrade:**
 - 11 fire extinguishers strategically allocated with a resident representative on each level of the 11-storied Blok 5 of Desa Mentari.
- **Community Involvement:**
 - 30+ local leaders and residents actively participated, increasing long-term accountability and overall fire awareness.
 - Raised awareness of proactive prevention measures to mitigate fires all together, safeguarding the local community.



The Involvement of Local Community Leaders



Engagement of Local Community Leaders

Final Reflections & Lessons Learned

Recommendations for Future Projects

- **Long-Term Maintenance Plan:**
 - Establish quarterly inspections to ensure all 11 extinguishers on each level of Blok 5 of Desa Mentari remain functional.
- **Scalability Strategy:**
 - Expand the initiative to other blocks of the Desa Mentari community, increasing fire safety and benefiting on cost effectiveness through economies of scale.
- **Sustainable Funding Approach:**
 - Combine corporate sponsorships + crowdfunding to reduce reliance on university sponsorship.
- **Community-Led Fire Safety Committee:**
 - Assign resident volunteers to monitor equipment and refresh training periodically.



Team behind the Safe Flame, Green Aim Project



Involvement of key stakeholders such as residents and leaders of the Desa Mentari Community